

1 Catalog: $|P| \leq 2$



$n = 3$ example

$\emptyset$

1

2

3

1,2

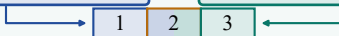
1,3

2,3

2 Directional certificates

$$P = \{1, 2\} \quad W_+(P) \leq \tau$$

$$Q = \{2, 3\} \quad W_-(Q) \leq \tau$$



3 Purchase $A = P \cup Q$



Charge once

Public

j_1

j_2 (shared)

j_3

(I_1, ε_1)

(I_2, ε_2)

(I_3, ε_3)

Reports

Y_{11}

Y_{12}

Y_{13}

$\vdots$

$\vdots$

$\vdots$

Y_{m1}

Y_{m2}

Y_{m3}

Cost split

$j_1 : 3$

$j_2 : 2$

$j_3 : 4$

$$C(P \cup Q) - C_0 = 3 + 2 + 4 = 9$$